PROJECT REPORT

VEHICLE INSURANCE FRAUD DETECTION

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ABOUT OUR PROJECT

Vehicle insurance fraud includes deceptive claims made after an accident, such as staged wrecks, bogus injuries, and phantom passengers. Detecting and combating these fraudulent tactics is critical for maintaining the insurance process's fairness and integrity. There are 33 fields and 15,420 records in the dataset utilized for analysis, giving a solid basis for spotting trends and anomalies connected to fraudulent actions.

OBJECTIVE

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The primary objective of this project is to develop an effective machine learning model utilizing a motor vehicle insurance dataset to accurately identify fraudulent insurance claims, ensuring improved fraud detection and enhancing the company's risk management capabilities.

As a team when we were discussing even before diving into analysis and test, we had a look at the data set and thought what are the factors that might define a case to be fraudulent.

With our basic realizations we thought factors like Policy Type, Type of vehicle, Presence of a witness, Age of vehicle and agent type might be some of the factors. With this our mind went ahead with our analysis to see if these were the factors or were there any more that were going to get added to this.

OVERVIEW OF THE DATASET

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Vehicle insurance fraud includes deceptive claims made after an accident, such as staged wrecks, bogus injuries, and phantom passengers. Detecting and combating these fraudulent tactics is critical for maintaining the insurance process's fairness and integrity. There are 33 fields and 15,420 records in the dataset utilized for analysis, giving a solid basis for spotting trends and anomalies connected to fraudulent actions.

Month	object
WeekOfMonth	int64
DayOfWeek	object
Make	object
AccidentArea	object
DayOfWeekClaimed	object
MonthClaimed	object
WeekOfMonthClaimed	int64
Sex	object
MaritalStatus	object
Age	int64
Fault	object
PolicyType	object
VehicleCategory	object
VehiclePrice	object
FraudFound_P	int64
PolicyNumber	int64
RepNumber	int64
Deductible	int64
DriverRating	int64

Days_Policy_Accident	object
Days_Policy_Claim	object
PastNumberOfClaims	object
AgeOfVehicle	object
AgeOfPolicyHolder	object
PoliceReportFiled	object
WitnessPresent	object
AgentType	object
NumberOfSuppliments	object
AddressChange_Claim	object
NumberOfCars	object
Year	int64
BasePolicy	object

SMART QUESTIONS

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These are a few questions that we would like to answer through our analysis.

- **Is there a correlation between the cost of a car and the likelihood of fraud?**
- **Is there a difference in the likelihood of fraud between accidents where someone is at fault and accidents where no one is at fault?**
- **Is there an effect of having a witness on the possibility of fraud in insurance claims?**

DATA PREPARATION

06

Even we being with our analysis let have a look at our data an our variables.
The first step would be cleaning our data and checking for Null Values.

Our dataset did not have any null values, but when we checked for the unique values present in each variable we saw that 'DayOfWeekClaimed' and 'MonthClaimed' have a '0' value in them which is not possible.

```
'WeekOfMonth': [5 3 2 4 1]

'DayOfWeek': ['Wednesday' 'Friday' 'Saturday' 'Monday' 'Tuesday' 'Sunday' 'Thursday']

'Make': ['Honda' 'Toyota' 'Ford' 'Mazda' 'Chevrolet' 'Pontiac' 'Accura' 'Dodge'
 'Mercury' 'Jaguar' 'Nisson' 'VW' 'Saab' 'Saturn' 'Porche' 'BMW' 'Mecedes'
 'Ferrari' 'Lexus']

'AccidentArea': ['Urban' 'Rural']

'DayOfWeekClaimed': ['Tuesday' 'Monday' 'Thursday' 'Friday' 'Wednesday' 'Saturday' 'Sunday'
 '0']

'MonthClaimed': ['Jan' 'Nov' 'Jul' 'Feb' 'Mar' 'Dec' 'Apr' 'Aug' 'May' 'Jun' 'Sep' 'Oct'
 '0']

'WeekOfMonthClaimed': [1 4 2 3 5]

'Sex': ['Female' 'Male']

'MaritalStatus': ['Single' 'Married' 'Widow' 'Divorced']

'Age': [21 34 47 65 27 20 36  0 30 42 71 52 28 61 38 41 32 40 63 31 45 60 39 55
 35 44 72 29 37 59 49 50 26 48 64 33 74 23 25 56 16 68 18 51 22 53 46 43
 57 54 69 67 19 78 77 75 80 58 73 24 76 62 79 70 17 66]

'Fault': ['Policy Holder' 'Third Party']

'PolicyType': ['Sport - Liability' 'Sport - Collision' 'Sedan - Liability'
 'Utility - All Perils' 'Sedan - All Perils' 'Sedan - Collision'
```

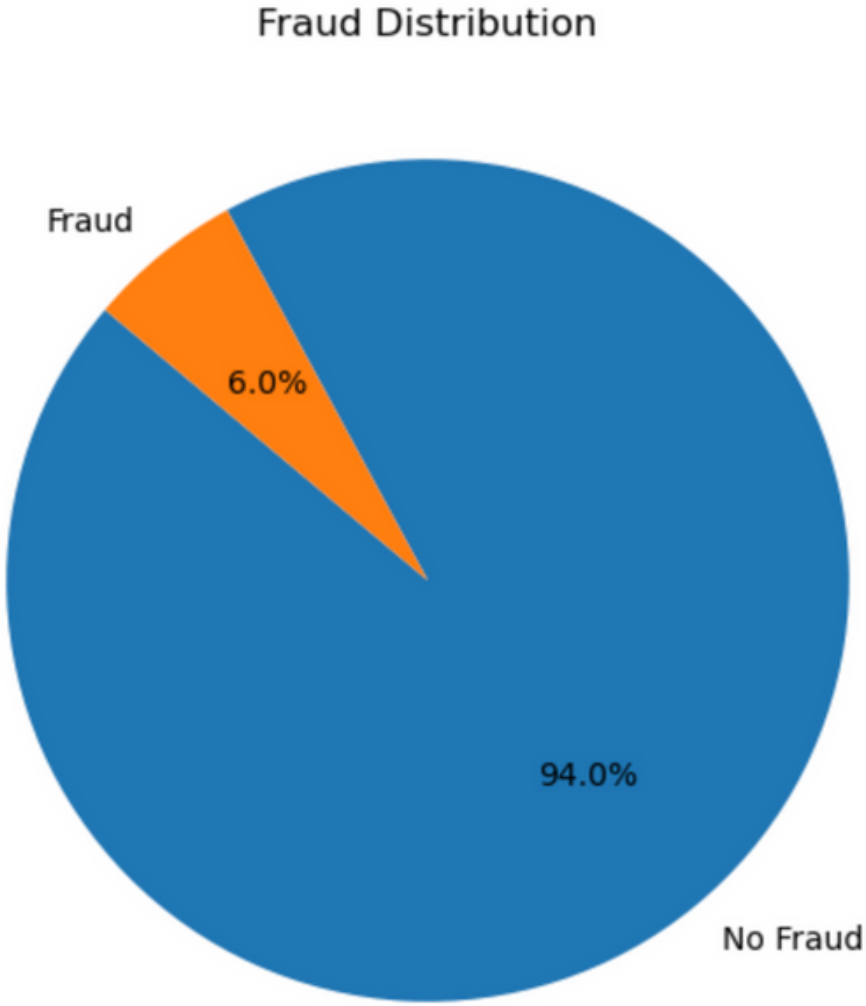
DATA PREPARATION

07

On Furthur investigating we see that there is one row in which has the '0' value. We feel that this may cause an error in our analysis and hence we have removed that entire row.

```
DayOfWeekClaimed has  1  row(s) with a 0
MonthClaimed has  1  row(s) with a 0

     Month  WeekOfMonth DayOfWeek   Make AccidentArea DayOfWeekClaimed  \
1516   Jul            2    Monday  Honda        Rural                0

     MonthClaimed  WeekOfMonthClaimed   Sex MaritalStatus  ...  AgeOfVehicle  \
1516            0                   1  Male        Single  ...           new

     AgeOfPolicyHolder PoliceReportFiled WitnessPresent AgentType  \
1516          16 to 17                No             No  External

      NumberOfSuppliments  AddressChange_Claim  NumberOfCars  Year  BasePolicy
1516                 none            no change     1 vehicle  1994  All Perils

[1 rows x 33 columns]
     Month  WeekOfMonth DayOfWeek   Make AccidentArea DayOfWeekClaimed  \
1516   Jul            2    Monday  Honda        Rural                0

     MonthClaimed  WeekOfMonthClaimed   Sex MaritalStatus  ...  AgeOfVehicle  \
1516            0                   1  Male        Single  ...           new

     AgeOfPolicyHolder PoliceReportFiled WitnessPresent AgentType  \
1516          16 to 17                No             No  External

      NumberOfSuppliments  AddressChange_Claim  NumberOfCars  Year  BasePolicy
1516                 none            no change     1 vehicle  1994  All Perils

[1 rows x 33 columns]
```

The total number of data is now '**15419**'



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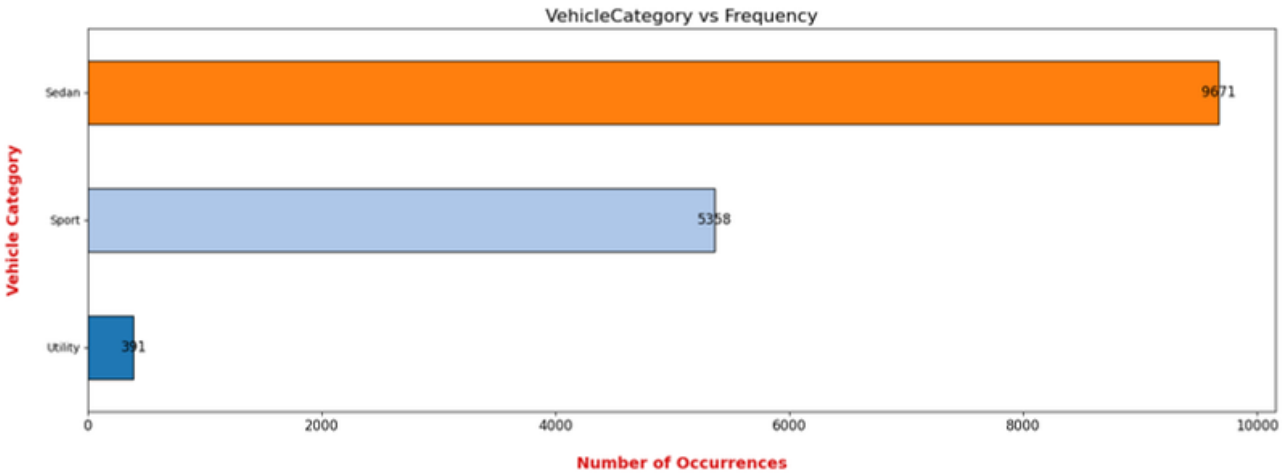
DATA VISUALIZATION

Our target variable is to check if the claim made is a fraudulent one or not. The above graph shows the overall distribution of cases in our dataset. We can clearly see that there are hardly any fraud cases compared to the non-fraud cases. For our initial analysis we will go ahed with this to see how indiviual variables affect this.

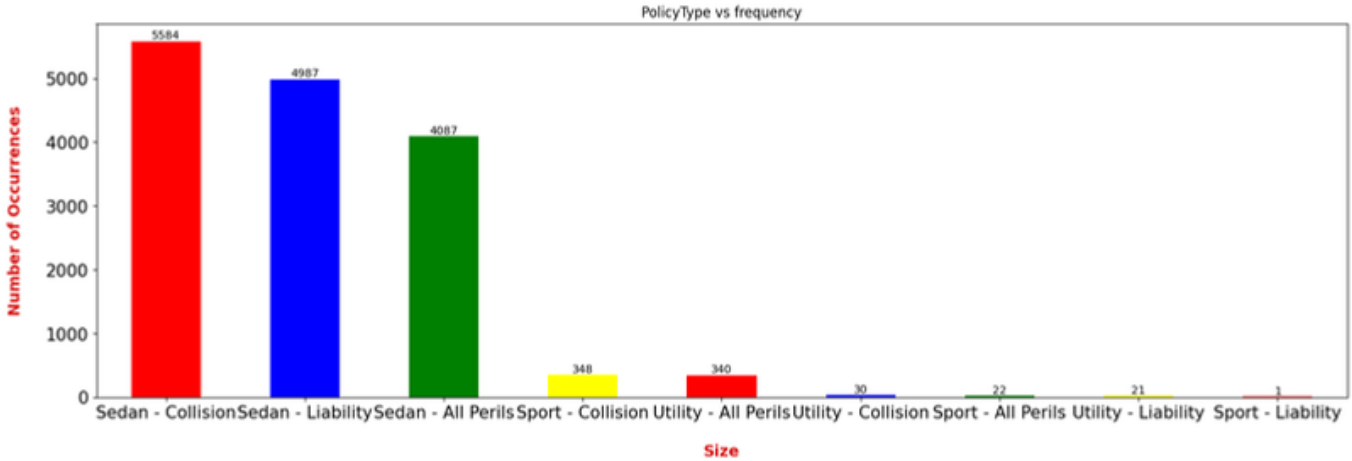
DATA VISUALIZATION

09

Let us now have a look at few variables individually!



There are more sedans at the top with 9671 total frequencies, followed by sport cars with 5358 frequency, and utility vehicles with the lowest frequency of 391.



Here is a graph that plots the information of the types of vehicles, their policy types, and their frequency in the dataset, and we can see that there are more sedans that have taken collision type of insurance, and we can see that there are more policies for sedans with collision type of insurance and sedans with liability insurance for about 4987 policies, and sedans with all perils come next to it, and all the other insurances have almost equal amount of policies with collision type of insurance.

DATA VISUALIZATION

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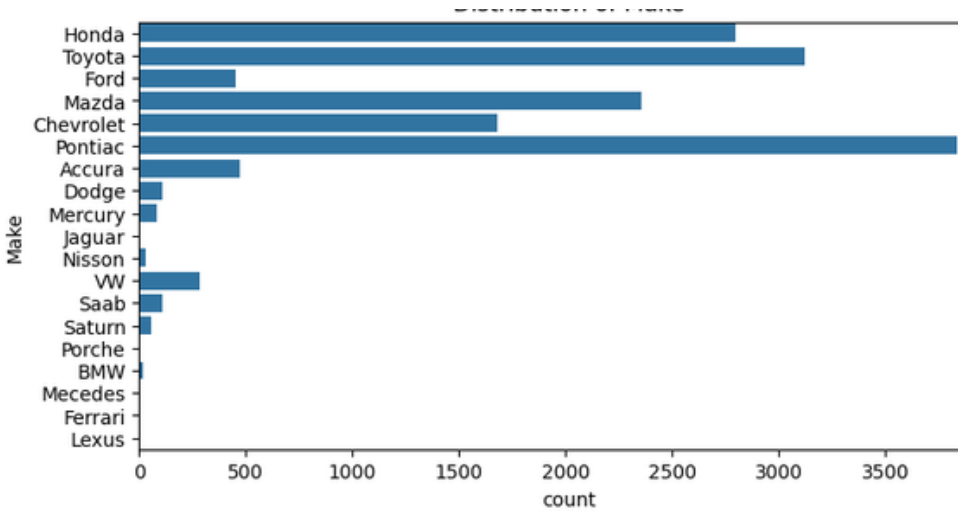
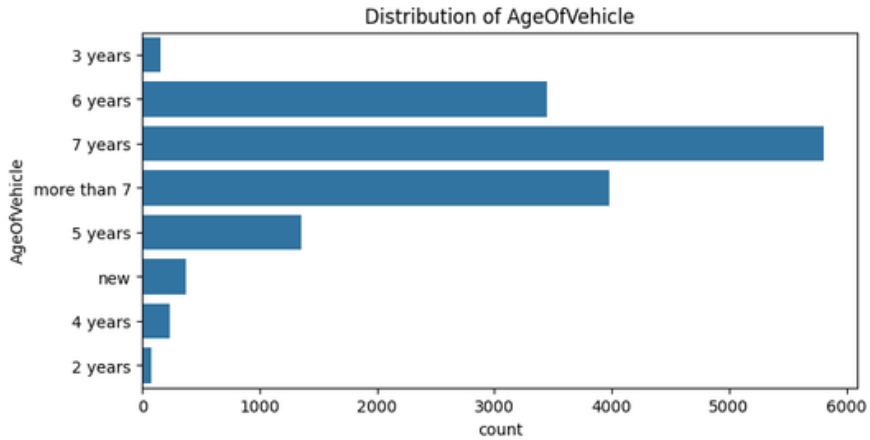
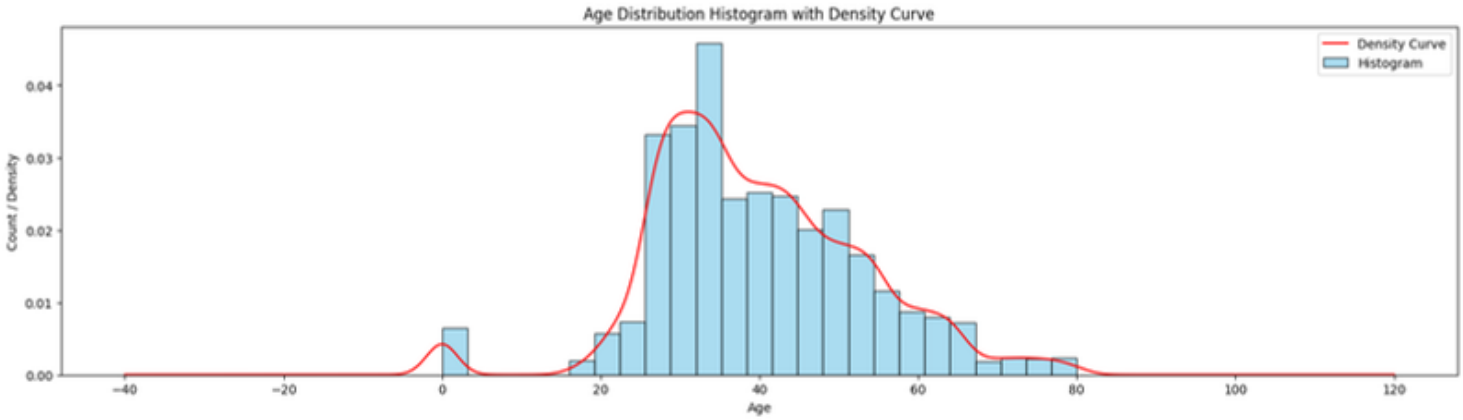


This graph shows the vehicle companies and the total number of policies, and we can see that Porsche has the most policies, followed by Toyota, with BMW having the fewest, and Mercedes, Ferrari, and Lexus have nearly equal numbers of policies for almost all of the policies.

This graph plots the age of the vehicle while the insurance time and their occurrences, and we can see that there are more policies for cars aged 7 years compared to cars aged more than 7 years, and we can see that there are fewer policies for cars aged 2 years.





This graph depicts the age distribution of consumers in the dataset, and we can see that most persons aged 20 to 60 are more likely to get insurance.

CORRELATION MATRIX

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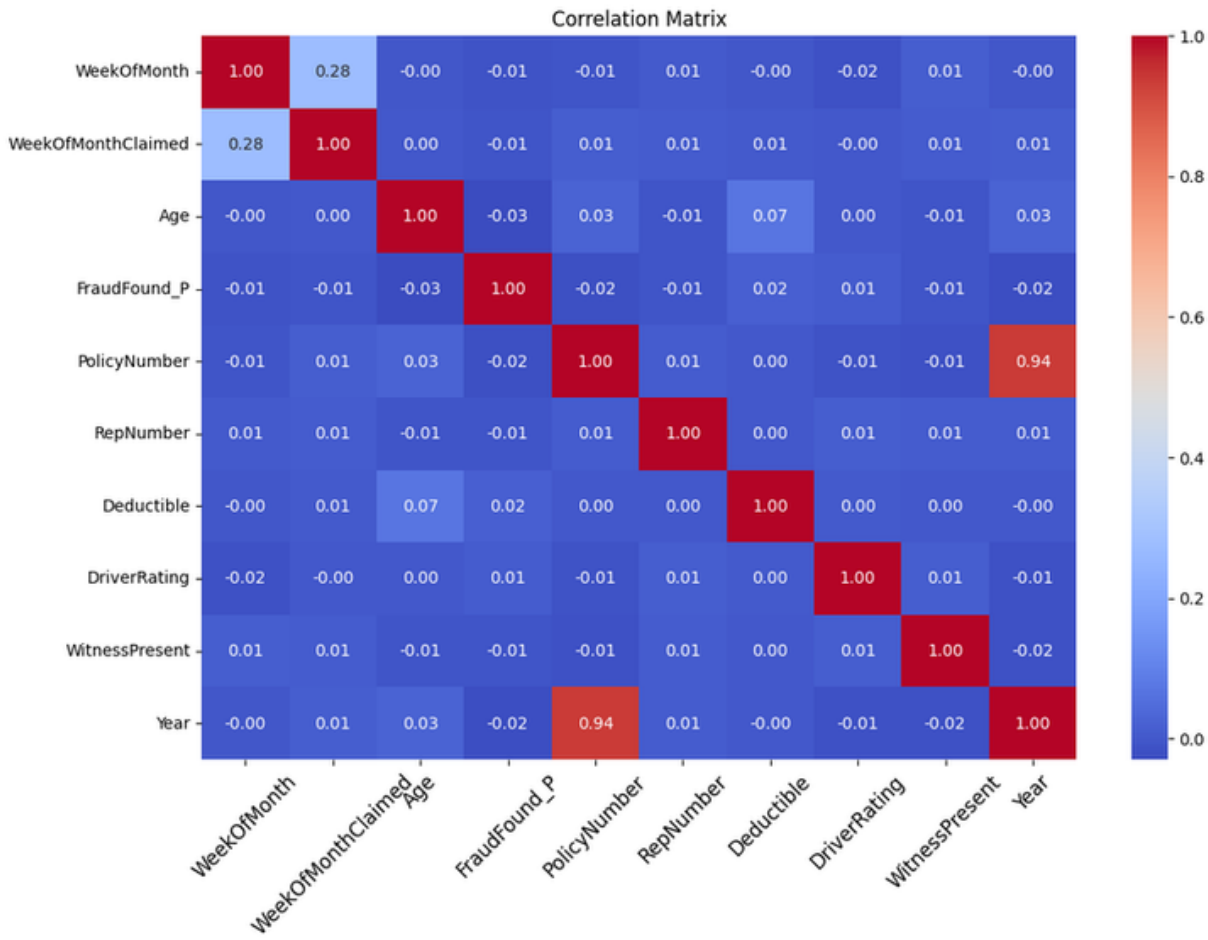
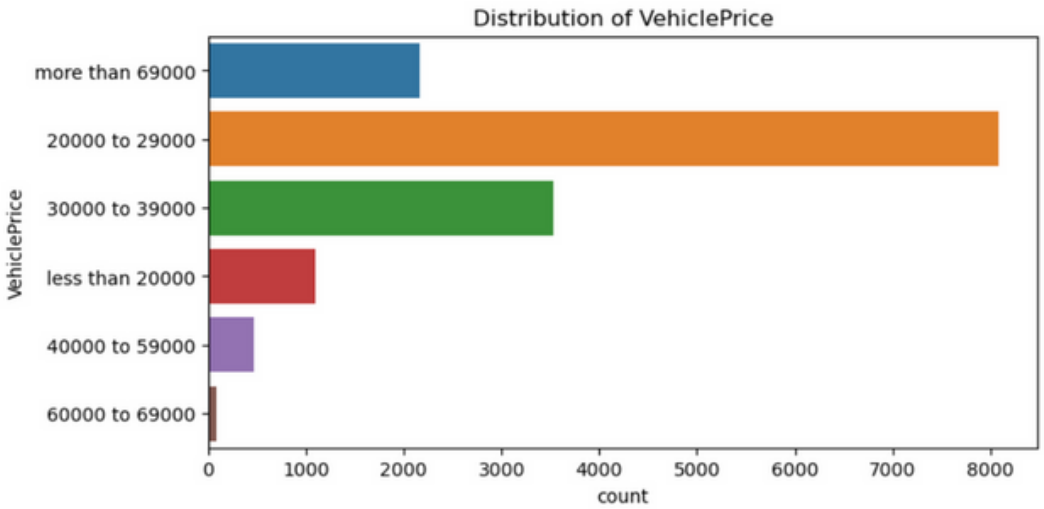


Before interpreting the correlations, there are some caveats to the data, From the matrix, there is a strong positive correlation between policy number and the year

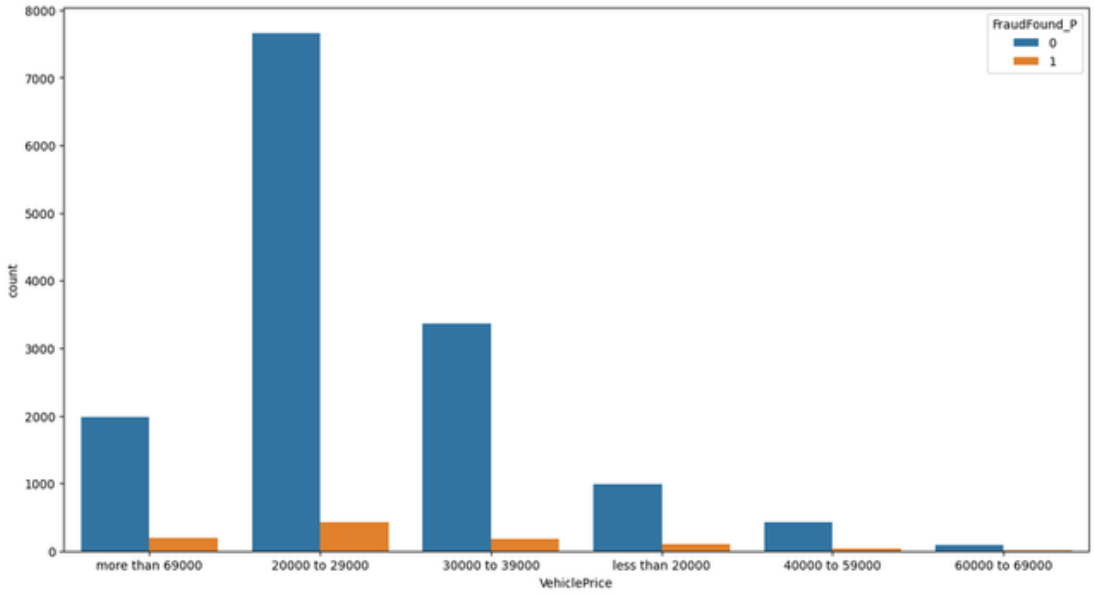
Questions:

- Is there a correlation between the cost of a car and the likelihood of fraud?
-
- Is there a difference in the likelihood of fraud between accidents where someone is at fault and accidents where no one is at fault?
-
- Is there an effect of having a witness on the possibility of fraud in insurance claims?

IS THERE A CORRELATION BETWEEN THE COST OF A CAR AND THE LIKELIHOOD OF FRAUD?



This is the graph that plots the fraud percentage in relation to the price range of the cars, and we can see that almost every car category has some fraud percentage, and cars priced between $20,000 and $29,999 have the highest amount of fraud rate during the time of insurance.



Here we have plotted a graph between the distribution of vehicle prices from the data and we can see that most of the vehicles have a price range of 20000 to 29000 dollars, some cars have a price range of 30000 to 39000 dollars, and many cars have a price range of more than 69000 dollars.

Chi-square statistic: 67.76
p-value: 2.9835975136646997e-13

The p-value is extremely modest, indicating that there is no significant link between car price and fraud identified, and the chi-square test findings strongly support the concept that there is no meaningful relationship between vehicle price and the chance of fraud. The percentage of false claims are the same across all car price groups.

IS THERE AN EFFECT OF HAVING A WITNESS ON THE POSSIBILITY OF FRAUD IN INSURANCE CLAIMS?

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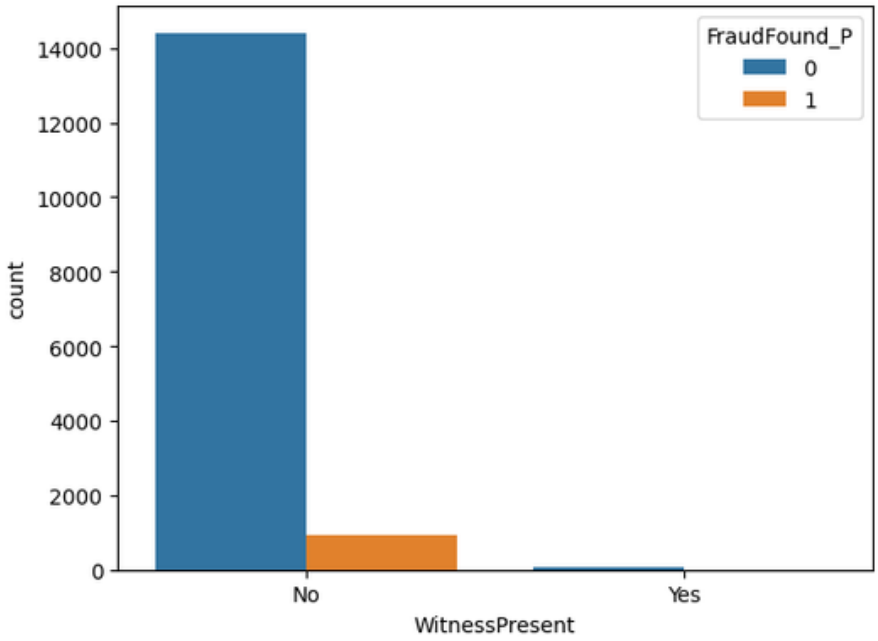

Here is the graph plotted between witness present on the x-axis and frequency on the y axis, we can say that most of the frauds have no witness present, though we can see there are some cases of witness present, and we can notice there is a slight notice in the frauds done and there are witness present in the results.

Chi-square statistic: 0.59
p-value: 0.438977726621

The chi-square statistic is 0.59, which is little, showing a minor difference between the actual and predicted values in the contingency table. The p-value is 0.44, which is high, indicating that the observed differences are not statistically significant. There is insufficient evidence to reject the null hypothesis that there is no link between the variables. In summary, the findings indicate that there is insufficient statistical evidence to infer that there is a substantial correlation between the existence of a witness and the occurrence of fraud. The proportions of false claims are the same whether there is or is not a witness.

STATISTICAL TESTS 14

IS THERE A SIGNIFICANT DIFFERENCE IN THE MEAN AGE BETWEEN INSURANCE CLAIMS WITH AND WITHOUT FRAUD, AND DOES THIS DIFFERENCE VARY BASED ON THE REPORTED DRIVER RATING?

Two-Way ANOVA Test: Mean Age in Insurance Claims with and without Fraud, Considering Driver Ratings

Objective:
The objective of this study was to investigate whether there is a significant difference in the mean age between insurance claims with and without fraud, and whether this difference varies based on the reported driver rating.

Hypotheses:

Null Hypothesis (H0):
There is no interaction effect between fraud (fraud vs. non-fraud) and driver ratings on the mean age.

Alternate Hypothesis (H1):
There is an interaction effect between fraud and driver ratings on the mean age.

Output:
The output of the two-way ANOVA test yielded the following results:

- **Fraud (FraudFound_P):**
  - Sum of squares (sum_sq): 2.482593e+03
  - Degrees of freedom (df): 1.0
  - F-statistic (F): 13.649244
  - P-value (PR(>F)): 0.000221

- **Driver Rating (DriverRating):**
  - Sum of squares (sum_sq): 1.769918e+02
  - Degrees of freedom (df): 3.0
  - F-statistic (F): 0.324366
  - P-value (PR(>F)): 0.807761

```
                                     sum_sq       df          F    PR(>F)
C(FraudFound_P)                  2.482593e+03      1.0  13.649244  0.000221
C(DriverRating)                  1.769918e+02      3.0   0.324366  0.807761
C(FraudFound_P):C(DriverRating)  1.067643e+03      3.0   1.956625  0.118163
Residual                         2.803212e+06  15412.0        NaN       NaN
P-value for Interaction Effect: 0.1181631806604882
Fail to reject the null hypothesis.
There is not enough evidence to claim a significant interaction effect.
```

- **Interaction Effect (FraudFound_P:DriverRating):**
  - Sum of squares (sum_sq): 1.067643e+03
  - Degrees of freedom (df): 3.0
  - F-statistic (F): 1.956625
  - P-value (PR(>F)): 0.118163

- **Residual:**
  - Sum of squares (sum_sq): 2.803212e+06
  - Degrees of freedom (df): 15412.0

P-value for Interaction Effect: 0.1181631806604882

Conclusion:

The p-value for the interaction effect between fraud and driver ratings on the mean age is 0.1181631806604882, which is greater than the commonly used significance level of 0.05. As a result, we fail to reject the null hypothesis. This implies that there is not enough evidence to claim a significant interaction effect between fraud and driver ratings on the mean age.

Interpretation:

1. Difference in Mean Age between Fraud and Non-Fraud Claims:
   - There is a significant difference in the mean age between insurance claims with and without fraud (p-value < 0.05). This suggests that age is a relevant factor in distinguishing between fraudulent and non-fraudulent claims.

2. Variation in Mean Age Based on Driver Rating:
   - There is no significant difference in the mean age across different levels of reported driver ratings (p-value > 0.05). This indicates that, overall, reported driver ratings do not contribute significantly to variations in mean age.

3. Interaction Effect between Fraud and Driver Rating on Mean Age:
   - The analysis does not provide enough evidence to support a significant interaction effect between fraud and driver ratings on the mean age (p-value > 0.05). This suggests that the impact of fraud on mean age does not vary based on the reported driver rating, and vice versa.

Overall Conclusion:
In summary, age appears to be a relevant factor in identifying fraudulent claims, as there is a significant difference in mean age between fraud and non-fraud claims. However, the reported driver rating does not contribute significantly to variations in mean age, and there is no significant interaction effect between fraud and driver ratings on mean age. Therefore, while age is a distinguishing factor, the reported driver rating does not appear to modify the relationship between age and fraud in a statistically significant manner.

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DATA MODELING

In data modeling, we harnessed the predictive capabilities of two distinct machine-learning models:

- **Decision Tree Classifier**
- **XGBoost (Extreme Gradient Boosting)**

DATA EXAMINATION AND MISSING VALUES:

In the preliminary phase of data preparation for model building, a thorough examination of the dataset was conducted to identify and address missing values. Fortunately, the dataset displayed a clean structure with no null values present in any of the columns. This absence of missing values ensures a solid foundation for the subsequent stages of machine learning model development.

Data Types Analysis:

The data types of each column were assessed to understand the nature of the variables and facilitate appropriate model-building strategies. The key columns include:

```
Month                 object
WeekOfMonth            int64
DayOfWeek             object
Make                  object
AccidentArea          object
DayOfWeekClaimed      object
MonthClaimed          object
WeekOfMonthClaimed     int64
Sex                   object
MaritalStatus         object
Age                    int64
Fault                 object
PolicyType            object
VehicleCategory       object
VehiclePrice          object
FraudFound_P           int64
PolicyNumber           int64
RepNumber              int64
Deductible             int64
DriverRating           int64
Days_Policy_Accident  object
Days_Policy_Claim     object
PastNumberOfClaims    object
AgeOfVehicle          object
AgeOfPolicyHolder     object
PoliceReportFiled     object
WitnessPresent        object
AgentType             object
NumberOfSuppliments   object
AddressChange_Claim   object
NumberOfCars          object
Year                   int64
BasePolicy            object
```

CATEGORICAL VARIABLES AND ONE-HOT ENCODING:

A crucial aspect of preparing data for model building involves handling categorical variables effectively. One such method is the application of one-hot encoding, which converts categorical variables into binary vectors, enabling machine learning models to interpret and utilize them more efficiently.

Noteworthy categorical variables and their corresponding one-hot encoding transformations include:

- NumberOfCars:
  - 3 to 4: uint8
  - 5 to 8: uint8
  - More than 8: uint8

- BasePolicy:
  - Collision: uint8
  - Liability: uint8
  - ... (additional categories)

The resultant DataFrame after one-hot encoding exhibits a concise and structured representation, providing a suitable input for machine learning models. The dataset now stands ready for the subsequent phases of model training, validation, and evaluation, with categorical variables appropriately encoded to enhance model performance.

```
Data types of each column:
Month                   object
WeekOfMonth              int64
DayOfWeek               object
Make                    object
AccidentArea            object
DayOfWeekClaimed        object
MonthClaimed            object
WeekOfMonthClaimed       int64
Sex                     object
MaritalStatus           object
Age                      int64
Fault                   object
PolicyType              object
VehicleCategory         object
VehiclePrice            object
FraudFound_P             int64
PolicyNumber             int64
RepNumber                int64
Deductible               int64
DriverRating             int64
Days_Policy_Accident    object
Days_Policy_Claim       object
PastNumberOfClaims      object
AgeOfVehicle            object
AgeOfPolicyHolder       object
PoliceReportFiled       object
WitnessPresent          object
AgentType               object
NumberOfSuppliments     object
AddressChange_Claim     object
NumberOfCars            object
Year                     int64
BasePolicy              object
dtype: object
```

```
Data types of each column:
WeekOfMonth                 int64
WeekOfMonthClaimed          int64
Age                         int64
FraudFound_P                int64
PolicyNumber                int64
                            ...
NumberOfCars_3 to 4         uint8
NumberOfCars_5 to 8         uint8
NumberOfCars_more than 8    uint8
BasePolicy_Collision        uint8
BasePolicy_Liability        uint8
Length: 125, dtype: object
```

DECISION TREE MODEL

The Decision Tree model presented in this analysis exhibits commendable performance with an accuracy of 91.5%. This indicates that the model is successful in correctly classifying instances into their respective categories. However, a comprehensive evaluation extends beyond accuracy, encompassing metrics such as AUC score, precision, recall, and the confusion matrix.

Model Metrics:

1. Accuracy: 91.5%

The accuracy of 91.5% denotes the proportion of correctly classified instances out of the total. It suggests that the Decision Tree model is proficient in making accurate predictions on the given dataset.

2. AUC Score: 64.96%

The AUC (Area Under the Curve) score of 64.96% is indicative of the model's ability to distinguish between classes. While the score is above random chance, further enhancements may be explored to boost discriminative power.

3. Precision/Recall Trade-off:

- Precision for fraudulent claims: 34%
- Recall for fraudulent claims: 35%

Precision measures the accuracy of positive predictions, indicating that 34% of identified fraudulent claims are accurate. Recall, on the other hand, gauges the model's ability to capture all instances of a particular class, with a 35% recall for fraudulent claims. The trade-off between precision and recall implies a delicate balance that may necessitate adjustments depending on the specific goals of the model.

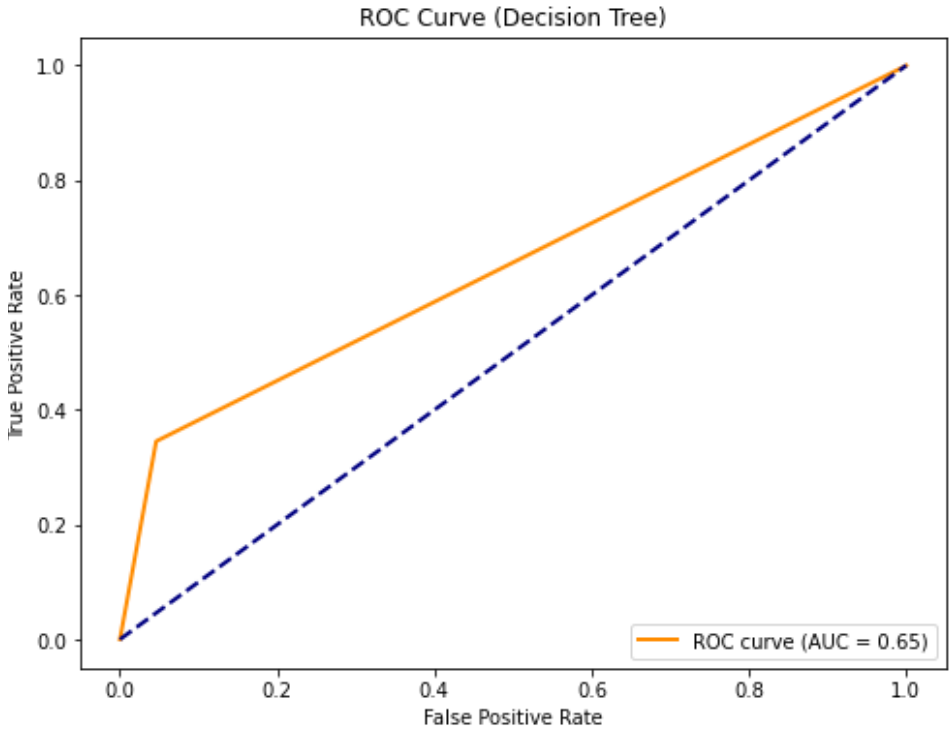
4. Confusion Matrix:

- True Negatives: 2754
- False Positives: 133
- False Negatives: 129
- True Positives: 68

The confusion matrix provides a detailed breakdown of the model's performance across different classes. Notably, the model achieved a high number of true negatives (2754), correctly identifying non-fraudulent claims. However, there were 133 false positives and 129 false negatives, suggesting areas where improvements can be made.

In summary, the Decision Tree model demonstrates robust overall performance with a notable accuracy level of 91.5%. Nevertheless, the trade-off observed between precision and recall highlights potential areas for further optimization. The balance between precision and recall may need to be adjusted based on the specific objectives and requirements of the application. Additionally, exploring alternative models or fine-tuning parameters could be beneficial to enhance specific aspects of predictive capability and achieve a more balanced performance across various metrics.

```
Accuracy with Decision Tree: 0.9150453955901426

AUC Score with Decision Tree: 0.6495545408350755

Classification Report with Decision Tree:
              precision    recall  f1-score   support

           0       0.96      0.95      0.95      2887
           1       0.34      0.35      0.34       197

    accuracy                           0.92      3084
   macro avg       0.65      0.65      0.65      3084
weighted avg       0.92      0.92      0.92      3084


Confusion Matrix with Decision Tree:
[[2754  133]
 [ 129   68]]
```



XGBOOST MODEL

XGBoost, renowned for its power in handling imbalanced datasets, emerges as an ideal candidate for this task. With its ensemble of decision trees, XGBoost excels in capturing intricate patterns within the data, a crucial aspect when discerning fraudulent claims from legitimate ones. The algorithm's boosting technique, which prioritizes the minority class, proves instrumental in addressing the inherent class imbalance common in insurance datasets. Furthermore, XGBoost's versatility in hyperparameter tuning allows us to fine-tune the model for optimal performance, making it a powerful and adaptive tool for enhancing fraud detection in motor vehicle insurance. The XGBoost model is by far the best performing model in our case.

Initial Model

The initial XGBoost model exhibited high accuracy and AUC score but struggled with the recall of fraudulent claims, indicating a challenge in identifying all actual fraud cases.

```
Accuracy with XGBoost: 0.958171206225681

AUC Score with XGBoost: 0.9760558006396608

Classification Report with XGBoost:
              precision    recall  f1-score   support

           0       0.96      1.00      0.98      2887
           1       0.85      0.42      0.56       197

    accuracy                           0.96      3084
   macro avg       0.91      0.71      0.77      3084
weighted avg       0.95      0.96      0.95      3084


Confusion Matrix with XGBoost:
[[2873   14]
 [ 115   82]]
```

XGBOOST MODEL WITH SMOTE

Since we had a highly imbalanced dataset, we tried oversampling the data using SMOTE. The introduction of SMOTE for oversampling led to a decline in performance, especially in terms of recall for fraudulent claims. The model struggled to identify a substantial number of actual fraud cases and gave worse results than the initial model. So, we decided to continue without oversampling the dataset.

```
Accuracy with XGBoost and SMOTE: 0.9471465629053177
AUC Score with XGBoost and SMOTE: 0.9571209289322518
Classification Report with XGBoost and SMOTE:
              precision    recall  f1-score   support

           0       0.95      0.99      0.97      2887
           1       0.74      0.27      0.39       197

    accuracy                           0.95      3084
   macro avg       0.84      0.63      0.68      3084
weighted avg       0.94      0.95      0.94      3084

Confusion Matrix with XGBoost and SMOTE:
[[2868   19]
 [ 144   53]]
```

HYPERPARAMETER TUNING OF XGBOOST MODEL

To enhance the performance of the XGBoost model, we conducted hyperparameter tuning using GridSearchCV. The objective was to find the combination of hyperparameters that maximizes the area under the Receiver Operating Characteristic (ROC) curve, as measured by the ROC AUC score.

Parameters Tuned:

Learning Rate: We explored values of 0.01, 0.1, and 0.2 to determine the impact on model performance.
Max Depth: Tuning the maximum depth of each tree in the ensemble, considering values of 3, 5, and 7.
Number of Estimators: Examining the effect of the number of boosting rounds, with options of 50, 100, and 200.
Subsample: Investigating the fraction of samples used in each boosting round, exploring values of 0.8 and 1.0.
Col sample by Tree: Determining the fraction of features used in each boosting round, considering values of 0.8 and 1.0.

Here are the results we got by tuning the hyperparameters:

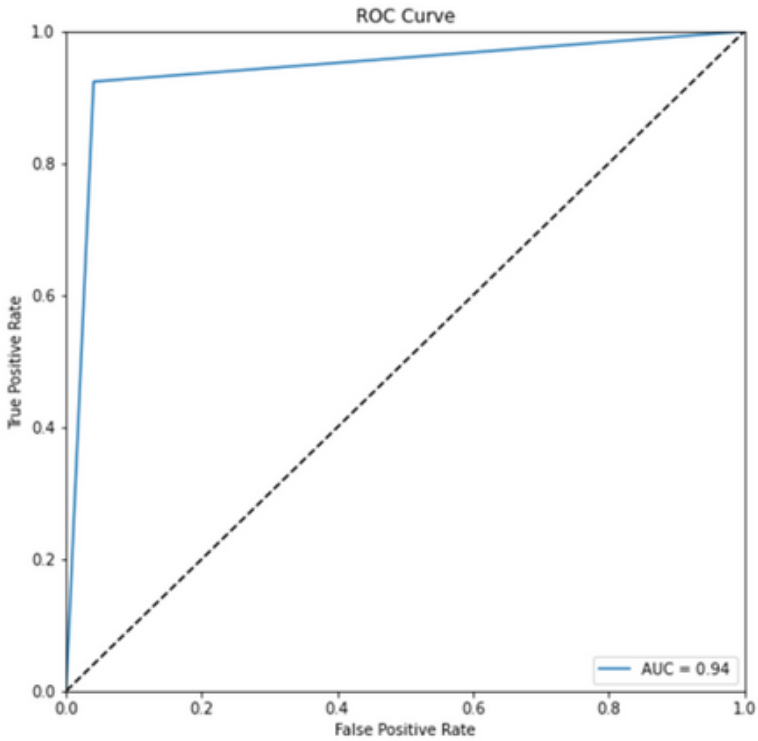
```
Accuracy with Best XGBoost Model: 0.9578469520103762
AUC Score with Best XGBoost Model: 0.9780180363927918
Classification Report with Best XGBoost Model:
              precision    recall  f1-score   support

           0       0.96      0.99      0.98      2887
           1       0.83      0.43      0.56       197

    accuracy                           0.96      3084
   macro avg       0.90      0.71      0.77      3084
weighted avg       0.95      0.96      0.95      3084

Confusion Matrix with Best XGBoost Model:
[[2870   17]
 [ 113   84]]
```

Performing hyperparameter tuning on the XGBoost model yielded slight improvements. The tuned model exhibited better precision, recall, and F1-score for fraudulent claims compared to the initial model.

FINAL XGBOOST MODEL WITH ADJUSTED THRESHOLD

In this final iteration, we applied a crucial refinement to the best XGBoost model by adjusting the classification threshold. The threshold was fine-tuned to 0.1, a strategic decision aimed at optimizing the balance between precision and recall. The model's performance metrics with the adjusted threshold are detailed below:

```
Accuracy of XGB with Adjusted Threshold: 0.9571984435797666
AUC Score of XGB with Adjusted Threshold: 0.9416656849626982
Classification Report of XGB with Adjusted Threshold:
              precision    recall  f1-score   support

           0       0.99      0.96      0.98      2887
           1       0.61      0.92      0.73       197

    accuracy                           0.96      3084
   macro avg       0.80      0.94      0.86      3084
weighted avg       0.97      0.96      0.96      3084

Confusion Matrix of XGB with Adjusted Threshold:
[[2770  117]
 [  15  182]]
```



The adjustment of the threshold is a critical step in customizing the model's behavior based on specific business needs and priorities. In fraud detection scenarios, especially for motor vehicle insurance, striking the right balance between different metrics becomes paramount.

Focus on Recall: In fraud detection, the consequences of missing a fraudulent claim (False Negative) can be substantial. Therefore, our emphasis was on maximizing recall to minimize the occurrence of False Negatives. By setting a lower threshold, the model becomes more sensitive, capturing a higher proportion of actual fraudulent cases.

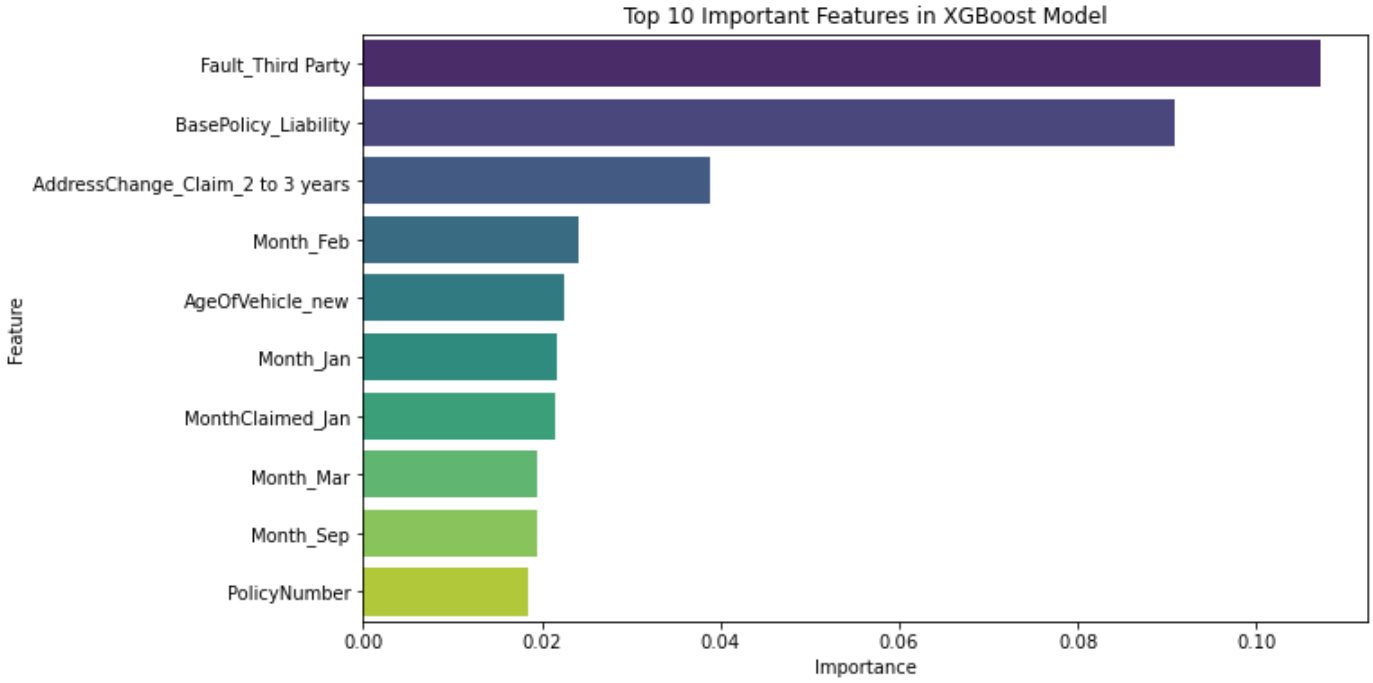
The trade-off with Precision: While prioritizing recall, precision (the ratio of correctly predicted positive observations to the total predicted positives) may experience a slight decline. This trade-off is acceptable in fraud detection, where a higher number of False Positives may occur (legitimate claims flagged as fraudulent), but the priority is to avoid missing potential fraud.

Impact on Confusion Matrix:
True Positives (TP): Increased due to the lower threshold, capturing more actual fraudulent claims.
False Negatives (FN): Decreased as a result of the increased sensitivity, minimizing the chances of missing fraudulent cases.
False Positives (FP): Experienced a increase, reflecting the compromise on precision for the sake of higher recall.
True Negatives (TN): Remains high, representing correctly identified legitimate claims.

Risk Mitigation Strategy: In fraud detection, the primary concern is often minimizing False Negatives. Adjusting the threshold to increase recall ensures a proactive approach to identifying potential fraud, even if it comes at the cost of a few more False Positives. This strategy aligns with the idea that it is more acceptable to investigate legitimate claims (False Positives) than to overlook actual instances of fraud (False Negatives).

Conclusion:
The XGBoost model with the adjusted threshold of 0.1 emerges as a robust solution for fraud detection in the motor vehicle insurance dataset. This approach ensures a high likelihood of capturing fraudulent claims, contributing to improved risk management and enhanced fraud detection capabilities for the insurance company.

FEATURE IMPORTANCE ANALYSIS: INSIGHTS FROM XGBOOST MODEL



In our pursuit of building an effective fraud detection model for motor vehicle insurance claims, we leveraged the power of XGBoost and conducted a comprehensive analysis of feature importance. The model, trained on our dataset, identified key features that significantly contribute to its decision-making process. The top 10 features, ranked by importance, are as follows:

- Fault_Third Party (10.71% Importance): This feature, representing fault attributed to a third party, emerged as the most critical in our model. Its high importance underscores its substantial influence on distinguishing between fraudulent and legitimate claims.

- BasePolicy_Liability (9.09% Importance): The type of base policy, specifically "Liability," holds significant weight in our model's predictions. Understanding the policy's fundamental nature proves crucial in detecting potential fraudulent activities.

- AddressChange_Claim_2 to 3 years (3.89% Importance): Changes in address, specifically in the timeframe of 2 to 3 years around the claim, contribute notably to the model's decision-making process.

- Month_Feb (2.40% Importance): The occurrence of claims in February exhibits a discernible impact on the model's predictions. The month of the claim is evidently influential in identifying suspicious patterns.

- AgeOfVehicle_new (2.26% Importance): The age of the vehicle, particularly the classification of "new," is a noteworthy contributor to our model's understanding of the data.

- Month_Jan (2.17% Importance): Similar to February, claims in January play a substantial role in the model's assessment of the data.

- MonthClaimed_Jan (2.14% Importance): Aligning with the importance of January as a claim occurrence month, the month of January in the claimed data reinforces its significance in the model's decision-making.

- Month_Mar (1.95% Importance): March, as a month of claim occurrence, demonstrates a discernible impact on the model's predictions.

- Month_Sep (1.95% Importance): Similar to March, claims occurring in September hold importance in our model's feature hierarchy.

- PolicyNumber (1.86% Importance): The policy number, while not the most influential, still plays a significant role in contributing to the model's understanding of the data.

Through our EDA and statistical analysis we did find a few results on the variables that influence whether the case is fraudulent or not. Age of the person holding the insurance seems to be an important factor whereas factors such as presence of witness, cost of car and driver rating do not seem to impact much.

From our modeling and analysis we found out that Fault is third Party and Base policy as liability are the two most influential factors that determine if a case is fraudulent or not.

When the model we built predicts a case as fraud, it is correct 61% of he time, and the model captures 92% of actual fraud cases.

Dataset Reference:

Kaggle - https://www.kaggle.com/datasets/shivamb/vehicle-claim-fraud-detection